

Beyond Price: A Bertrand–Hotelling Empirical Analysis of Differentiated Competition in the Traditional Bazaars of Xinjiang

Weichen Xu

*Suzhou Foreign Language School, Suzhou, China
xuweichen2027@outlook.com*

Abstract. This study aims to determine whether the traditional bazaars in Xinjiang are pure price-competitive markets or multi-dimensional systems where competition involves factors such as price, trust and space. Based on the Bertrand model of price competition and the Hotelling linear-city model of spatial competition, an integrated theoretical framework has been constructed, and its main parameters—price sensitivity (Δ), the proportion of trust- or service-oriented consumers (α), and the unit spatial cost (t)—have been operationalised through a structured questionnaire distributed to 90 consumers and a parallel merchant sample. Composite indices of price sensitivity, trust and service orientation, space cost, search behaviour and first-seen purchase tendency are constructed from Likert items and included as explanatory variables in ordinary least squares regressions, with a baseline specification and education- and age-based subgroup specifications. Based on the above analysis, the main reasons why people choose bazaar vendors over supermarkets are that they are more price-sensitive and price-conscious; in addition, factors such as trust and service-orientation and impulsive buying behaviour are also responsible, and these effects vary with education level and age group. Perceived space cost and explicit comparison-search behaviour do not offer independent explanations. The above results indicate that traditional bazaars do not conform to the classic Bertrand hypothesis of price convergence for homogeneous goods and are better understood as differentiated, socially embedded markets where relational and behavioural mechanisms supplement the price mechanism.

Keywords: traditional bazaar, price competition, Hotelling model, consumer behavior, social embeddedness

1. Introduction

1.1. Background

In the context of the ongoing modernization and digitization of the global retail system, traditional market forms are facing significant transformational pressures. The rise of supermarkets, e-commerce and the platform economy has led to a significant increase in the efficiency of goods distribution, while also changing the decision-making logic of consumers. According to the World

Bank [1], e-commerce penetration in developing countries continues to rise, with more and more consumers preferring to purchase standardized goods through online channels.

However, parallel to this trend, traditional markets remain highly active in many regions, especially in developing economies. Studies in rural China have shown that traditional markets not only serve the function of commodity distribution, but also play an important role in the sale of agricultural products, informal employment and social interactions [2]. In addition, statistics from the National Bureau of Statistics of China [3] show that a large percentage of the rural population still relies on offline markets for their daily consumption and production activities.

The above phenomena are also particularly noticeable in the minority area of Xinjiang. The old city and its markets are still bustling places for trade today, and they also host many cultural events. Its operation is related to the people's religion, ethnic groups and local social structures [4]. The three elements of economy, society and culture together have taken a different path of development compared with the urban commercial system under modernisation.

Therefore, a new problem has slowly arisen—why are the old bazaars still in use in the age of modern trade? Does the present system of internal competition conform to the "price-led market" hypothesis in classical economics? This is the problem of actual life and the foundation for this study.

1.2. Research aims and questions

Traditional microeconomic models usually understand market competition as a price-centered game process. For example, under the assumption of homogeneous products, the theory of price competition predicts that firms will compete for market share by continually lowering prices, thus converging prices to the level of marginal cost. However, this theory has obvious limitations in explaining traditional bazaars in reality.

Existing studies have suggested that bazaar transactions are not only affected by price, but also by the significant role of trust relationships, social networks and cultural habits [5]. At the same time, spatial structure and location conditions also influence consumer choice behavior to a great extent [2]. In addition, behavioral economics further states that consumers are not completely rational, but make decisions under the constraints of limited information and psychological preferences [6].

Therefore, a single price-competition model is not suitable for the complex behavioural patterns in real bazaars. The environment in the Xinjiang bazaar has a relatively strong social embeddedness, and thus the behaviours of consumers and merchants may be more complex in many ways. Based on the above, the main research questions of this paper are as follows.

RQ1—What are the main drivers of the competition mechanism in the Xinjiang bazaar, price, or are there other reasons?

RQ2—How are price sensitivity, trust, and space related to consumer choice and merchant strategy?

RQ3—Are the traditional bazaars different from what classical economics considers a "purely price-competitive market"?

Based on the answers to the above questions, this paper will explore the working mechanism of traditional bazaars in both theory and practice.

2. Literature review

2.1. The embeddedness of traditional bazaars: the social basis of economic behavior

Research has generally shown that the traditional market is not only a place to buy and sell at a reasonable price but is also a type of economic organisation with a long history. According to the theory of embeddedness proposed by Karl Polanyi, economic activities are not autonomous but are closely related to society and constrained by institutions [7]. Social capital theory also provides an explanation for the above relationship of exchange. Pierre Bourdieu [8] was the first to propose that one's social group and confidence in others could be viewed as a form of economic capital affecting economic behaviour. In a market, familiar people or people one has spoken to often have less risk and are therefore chosen.

The theory has been well-supported in rural China to date. Lu Yilong [9] believes that the rural market is a place for the exchange of goods; at the same time, it also serves as a hub for social interaction and the manifestation of cultural identity, and its construction and changes reflect the increasing reliance of rural society on the market. This integration is relatively weak in ethnic-minority areas. Meng Qi [4] found that the Uyghur "Bazaar" serves economic purposes, but is also closely related to religious morality and social ties, so its commercial activities are restricted by culture. Therefore, the traditional bazaar should be understood as a "socially embedded market" that is not only governed by price but also by trust, culture and social relations.

2.2. Bazaars under the impact of modern commerce

With the development of the modern retail system, scholars have begun to investigate how the traditional bazaar is being changed. New institutional economics and consumer behaviour research have offered new explanations. Kotler and Keller [10] have shown that the reasons for people's choices are not only limited to price, but also include psychological, cultural and social factors. Oliver Williamson [11] thinks that for high-uncertainty or high-logistics-cost transactions, off-the-shelf markets are more efficient.

Sun Jian [5] found from empirical research that the choice of "bazaars" or "supermarkets" by rural residents is not solely based on price and quality, but also on cultural habits and social interaction; thus, the two types of markets have had a complementary effect. Yang and Liu [2] have also pointed out that, in terms of the distribution of agricultural products and supplementary employment, rural bazaars are still indispensable for older age groups in society. Therefore, the traditional market system has not disappeared in the modern economy; rather, it has coexisted with modern trade for a long time through functional differentiation and institutional adjustment.

From a development perspective, traditional bazaars are in a dynamic process of transformation. While urbanization and population mobility have led to the decline of some bazaars, bazaar functions are simultaneously expanding. Tiemulun [12] found that some rural bazaars have gradually transformed from a single trading place into a multifunctional space integrating culture, entertainment and information exchange. This transformation can be explained by the theory of "structural transformation" in development economics, that is, traditional economic forms do not disappear in the process of modernization, but are regenerated through functional restructuring.

2.3. Spatial structure and market organization

From the perspective of spatial economics, market competition is reflected not only in the price dimension, but also in the spatial dimension. The classic Hotelling [13] model points out that firms' choice of location in space affects the pattern of market competition. Meanwhile, there is an interaction between space and social relations. Henri Lefebvre [14] put forward the theory of the "production of space", that is, that space is not only a physical existence, but also a product of social relations. In bazaars, spatial location is often combined with trust networks and social interactions to form a complex competitive structure. In studies of rural bazaars, Yang and Liu [2] found that bazaars show point, line and block distribution, and their spatial structure is closely related to transportation conditions and population distribution. In addition, space not only affects consumer mobility costs, but also influences information acquisition and comparison behavior. However, most existing studies remain at the level of empirical description and lack an analytical framework that integrates spatial theory with price-competition models.

2.4. Multidimensional mechanisms of consumer behavior

Traditional microeconomics assumes that consumers are perfectly rational, but behavioural economics argues otherwise. Kahneman and Tversky [6] proposed in Prospect Theory that a person's selection is based on psychology and the perception of risk. George Stigler [15] put forward the idea that consumers consider the cost of searching in terms of the benefits gained from such a search; more recently, studies have expanded on this to demonstrate that external search effort varies systematically among different products and consumers [16]. In the market, consumers are often willing to make better decisions by comparing the contents of various stalls, and this is a typical form of "limited rationality".

Many kinds of decisions are made at different times in the market. Sun Jian [5] found that the consumers of the bazaars are not only concerned with prices, but are also affected by familiarity and cultural customs. Ren Zhixia [17] added that the functions of consumption and entertainment in bazaars are both attractive; they offer "experiences to participate in" rather than just goods for sale. In line with the theory of planned behaviour [18], these attitudinal, normative and perceived-control factors jointly drive consumers' intentions. In this paper, the theory of search behaviour is also applied to study how bazaars are used. Therefore, bazaar consumption behaviour should be understood as the result of combined factors of price, trust mechanisms and search behaviour, rather than being driven by price alone.

2.5. Literature review and research gaps

Most existing research has been carried out from a sociological and anthropological perspective, focusing individually on price, trust and space; these studies have predominantly employed qualitative methods or broad-scale data and have not established an integrated system for quantifying the impact of these factors at a micro level. In response to the aforementioned deficiencies, this paper integrates Bertrand price competition and Hotelling spatial competition with the social-embeddedness perspective, and operationalizes the key parameters (Δ , α , t) via questionnaire-based composite indices; thus, the bazaar is treated as a multi-dimensional system in which price, relational and spatial mechanisms operate concurrently.

3. Methodology

3.1. Research design and theoretical framework

This study adopts a quantitative empirical approach based on a questionnaire survey. It integrates the Bertrand price-competition model [19] and the Hotelling linear-spatial-competition (linear city) model [13] in microeconomics. The aim is to examine how different competition mechanisms shape consumers' purchasing choices and merchants' pricing and location-selection behavior in the context of offline bazaar purchasing in Xinjiang. Because direct observation of merchants' strategic interactions in bazaars is difficult to implement—given limitations in data accessibility, continuous tracking costs and privacy and ethical constraints—this study instead uses survey-based behavioral and cognitive responses from consumers and merchants to proxy key parameters in the relevant game models and to link the theoretical framework with observed market outcomes.

At the theoretical level, the Bertrand model is mainly used to explain how price differences affect market-share allocation under homogeneous-product conditions, while the Hotelling linear-spatial-competition (linear city) model is used to characterize the role of spatial location and the spatial cost parameter (t) in competition. This paper combines the two to construct an analytical framework that incorporates both price competition and spatial competition, in order to better capture the institutional characteristics and operating mechanisms of traditional bazaars, such as multi-vendor co-location, offline search frictions, spatial clustering and persistent price dispersion.

3.2. Questionnaire design and variable construction

The questionnaire data are organized as follows. Based on the general directions for multi-item construct measurement [20], the instrument was developed. The two sections of the questionnaire are a consumer questionnaire and a merchant questionnaire; they separately collect demand-side preferences and choice frictions (e.g., price sensitivity, types of consumer preferences, spatial behaviour and search costs) and supply-side strategic perceptions (e.g., pricing responses and location-based competition). Most of the attitude items are in the form of a five-point Likert scale (1 = strongly disagree; 5 = strongly agree). In line with the above theory, the main parameters of the game model in this paper have been systematically mapped to questionnaire items, as shown in Table 1. Each parameter is a composite index consisting of multiple components, and reverse coding of certain items may be required.

Table 1. Mapping of theoretical model parameters to questionnaire variables

Theoretical model	Parameter	Economic meaning	Questionnaire module	Composite variable
Bertrand	Δ	Consumer tolerance threshold for price differences	A	Price Sensitivity
Bertrand	α	Share of price-oriented vs service-oriented consumers	B	Preference Type
Bertrand	pA, pB	Merchant pricing strategy and response	D	Price Reaction
Hotelling	t	Consumer spatial cost	C	Space Cost
Hotelling	x_A	Stall spatial position	E	Location Sensitivity

3.3. Measurement of consumer behavioral variables

3.3.1. Price sensitivity (Δ)

Price sensitivity refers to the smallest change in price needed for a consumer to switch from choosing one good to another, and it is denoted as Δ in the Bertrand model. Many items in this paper are used to measure consumers' reactions to changes in price under different circumstances, such as (1) whether a small price difference can lead to a change in purchase decisions; (2) how consumers trade off price with other factors such as trust and other service attributes when these factors conflict; and (3) whether an unexpected price drop would alter a planned purchase at a well-known stall. Some items are phrased in reverse to control for response sets and have been reverse-coded before aggregation.

Reverse coding is used for the relevant items in data processing to ensure directionality, and then an average is taken to construct a price-sensitivity index, following the general composite-index formula (1) employed throughout this paper for all attitude composite indices.

$$Index_i = \frac{1}{k} \sum_{j=1}^k x_{ij} \quad (1)$$

where Index denotes any composite index (e.g., PriceSens, Trust, SpaceCost), i indexes individual respondents, and k is the number of items contributing to that index after reverse-coding. Higher values of this index indicate greater price sensitivity; under the model assumption, this implies a lower price-tolerance threshold Δ , noting that Δ is conceptual and the index serves as its empirical proxy rather than a directly calculated parameter.

3.3.2. Consumer preference type (α)

Consumer preference type is used to differentiate price-oriented consumers from trust- or service-oriented consumers, corresponding to α , the share (proportion) of a given consumer type in the Bertrand model. The questionnaire measures the trade-offs between price and non-price factors by setting up questions on the degree of price preference, the trust brought by familiarity with vendors and the impact of service quality on consumers' purchase decisions. By averaging the scores of the relevant items (after reverse-coding where applicable), the overall preference orientation of consumers in the sample can be assessed, thereby reflecting the relative proportions of different consumer types in the sample.

3.3.3. Spatial behavior and search cost (t)

The spatial-behavior variable is used to measure consumers' perceived spatial cost and comparison-related movement frictions within the bazaar, and it serves as a proxy for the spatial cost parameter t in the Hotelling model. Relevant items measure the preference for stalls near the entrance, the willingness to increase walking distance in exchange for a lower price, the importance of stall location for the purchase decision and the willingness to compare multiple stalls; in this paper, only the items that directly capture distance-related cost and avoidance (rather than evaluative outcomes) are used to construct the space-cost index.

In the index construction, the space-cost index is formed by synthesizing the distance-sensitivity item and its reverse-coded counterpart (reverse-coded prior to aggregation). Higher values of the

index indicate that consumers are more sensitive to spatial distance, that is, a greater utility loss per unit of distance.

3.4. Measurement of merchant behavioral variables

The merchant questionnaire is mainly used to analyze the manifestation of price competition and spatial competition on the supply side. The price-response items measure merchants' attention to competitors during pricing, their flexibility in adjusting prices and their degree of price rigidity, thereby serving as empirical proxies for the price-response function implied by Bertrand-type competition.

Meanwhile, merchants' subjective judgments of consumers' price sensitivity and preference orientation (price-oriented vs. trust- or service-oriented) can be compared with and validated against the consumer questionnaire data to assess potential information asymmetry or strategy mismatch, using the mean difference and correlation between merchant ratings and consumer self-reports as identification criteria. In addition, the items related to stall location are used to approximate the perceived effect of spatial location on consumer visit and purchase propensity, reflecting the perceived returns to location choice in the Hotelling framework.

3.5. Econometric model

The core dependent (outcome) variable of this paper is consumers' propensity to choose bazaar vendors (stall vendors) over supermarkets for purchasing products, denoted as *ChooseVendor*. This variable is derived from the consumer questionnaire item on overall purchase propensity and is measured on a five-point Likert scale, where higher values indicate a stronger tendency to purchase from bazaar vendors (1 = strongly prefer supermarkets; 5 = strongly prefer bazaar vendors). Since this variable is an ordered attitudinal measure, it is treated as approximately continuous in this paper and estimated using a linear regression model; this OLS specification is a common approximation for Likert-type outcomes, and robustness can be assessed using rank-based correlations or ordered-response models such as those reviewed by McFadden [21] if needed.

3.6. Linear regression model

Based on the above variable constructs, this paper uses ordinary least squares (OLS) [22] to estimate the following linear regression model.

$$\begin{aligned} ChooseVendor = \beta_0 + \beta_1 PriceSens + \beta_2 Trust + \\ \beta_3 SpaceCost + \beta_4 Search + \beta_5 FirstSeen + \varepsilon \end{aligned} \quad (2)$$

in which ε is a random error term that reflects individual-level preference differences not explicitly captured by the model. To improve coefficient comparability and facilitate interpretation, all continuous explanatory variables are standardized before regression. The estimation results are mainly used to test the expected coefficient signs and assess the relative contributions of the mechanism blocks—price competition (Price Sensitivity, Search Behavior), non-price orientation (Trust and Service Orientation), spatial frictions (Space Cost) and behavioral control (First-seen Purchase)—in explaining consumers' propensity to choose bazaar vendors. Table 2 reports the definition of each variable and its expected sign.

Table 2. Definition of regression variables and expected signs

Variable	Meaning	Source	Theoretical mapping	Expected sign
ChooseVendor	Tendency to choose bazaar vendors	Consumer questionnaire	—	—
Δ (PriceSens)	Price sensitivity	A1–A3, B1	Bertrand	+
Trust	Trust and service orientation	B2–B3	Bertrand (α)	+
SpaceCost	Perceived spatial cost	C1–C3, C6	Hotelling (t)	$\pm$
Search	Comparison and search intensity	C4	Hotelling	+
FirstSeen	Impulsive purchasing	C5	Behavioral control	–

4. Results

4.1. Demographic characterization

In terms of gender distribution, 61.11% of the respondents in the sample were male and 38.89% were female, with the mean value of the proportion of females being 0.39. The overall sample showed the structural characteristic that there were slightly more males than females. This result shows that the consumer behavior of the bazaar reflected in this study is to a certain extent more skewed towards the perspective of male consumers, but female consumers still account for an important proportion of nearly 40%, which provides a basis for the subsequent analysis of gender differences.

The four age groups in the age variable are less than 18 years old (1), 19–36 years old (2), 37–54 years old (3), and 55–72 years old (4). The mean is 3.17; therefore, the sample is relatively skewed towards middle-aged and older consumers, with 83.33 per cent of the respondents being aged 37 and over, and only 16.67 per cent being under 36. Based on the above analysis, it can be inferred that the consumption data in this study mainly reflect the consumption decision logic of middle-aged and elderly consumers in the bazaar setting, rather than consumption decision patterns led by younger generations or students.

The educational attainment variable has five levels—primary and junior high school, senior high school, college and technical secondary school, undergraduate, and graduate and above; a small number of these values may be doctoral degrees or higher. The average is 3.11, and this corresponds to the category of "college-bachelor's degree". Based on the consumer questionnaire, the distribution of educational attainment is as follows—graduate students 25.56%, college 24.44%, high school 22.22%, elementary and junior high school 18.89%, undergraduate 3.33%, and other (possibly Ph.D.) 5.56%. Most of the respondents (around 53.33%) have a college degree or higher, and the proportion of graduate students is relatively large.

4.2. Scale analysis

Scale analysis was conducted to verify that the price-sensitivity (A) and loyalty and trust (B) dimensions capture distinct constructs. The correlation between the A-index and the B-index is near zero ($r \approx -0.03$), supporting their treatment as separable rather than as a single latent factor — empirically establishing the co-existence of "price sensitivity" and "relational loyalty" in bazaar consumption. For the spatial-behavior dimension (C), item-level means indicate active comparison rather than convenience-seeking—walking longer for a lower price (mean = 3.96) and comparing multiple stalls before buying (mean = 3.87) score notably higher than convenience-related items

(e.g., shopping at the entrance for time-saving, mean = 2.93). Overall, the pattern is consistent with active search-and-compare behavior layered on top of underlying price sensitivity, rather than convenience-oriented purchasing.

5. Discussion

5.1. Price and loyalty

Trust and "soft services" appear to reduce price elasticity, creating local monopoly effects. In the loyalty dimension, the overall mean value of module B is higher, suggesting that trust relationships and micro-services (e.g., attitude, familiarity, neatness) are associated with real economic value in the bazaar. More specifically, the data support the interpretation that higher trust items correlate with higher vendor choice propensity to accept a premium price. In addition, the positive correlation between A2 and B3 ($r = 0.27$) suggests that even consumers who are skewed towards trusting supermarkets also place a high value on the service experience. From a modeling perspective, this finding is consistent with the idea that service and trust reduce consumers' price sensitivity for trusted vendors. As a result, even if prices are slightly higher, customers may not be completely lost, and competition in the market could shift from 'total loss' to 'partial loss'. This pattern is consistent with what one might expect under relationship-based market power.

Therefore, trust in the market needs to address psychological deficiencies and serve as an economic tool to stabilise demand and build a healthy profit-making environment; it will help curb price competition.

Economically speaking, the results do not meet the requirements of the classic Bertrand model of "homogeneous products + perfect rationality + no switching costs", which predicts that "prices will reach marginal cost". Therefore, according to the above analysis, the new equilibrium satisfies the condition for a Bertrand model with product differentiation, $\delta > 0$. Under this model, the firm can maintain an equilibrium price above its cost without losing all demand, as long as there is a positive utility gain for consumers from services, trust or quality. The data in this study are consistent with product differentiation, but they do not directly estimate δ or observe prices and costs.

5.2. Spatial behavior

Space and search costs exist, but perceived spatial frictions appear limited and do not constitute the main reason for higher equilibrium prices. In the spatial-behavior (C) module, consumers as a whole show a strong willingness to compare, but are not highly sensitive to physical distance. This implies that in the overall sample, the spatial and search cost t is positive but close to 0. Theoretically, this result does not align with an interpretation in which spatial frictions are the primary driver of higher prices. Instead, it is closer to a "weak spatial friction" environment in which consumers can generally compare prices, but limited attention and physical costs still remain. One thing to note is that the actual value of t may be higher in the older age groups (e.g., due to physical constraints). However, in the overall sample, this effect is not sufficient to be a central mechanism for raising the equilibrium price in the market.

5.3. Comparison and synthesis of an improved model

Based on the above results, a simplified improved model can be constructed. As an illustrative assumption only, let the two types of sellers be market vendor A and supermarket B, whose marginal

costs satisfy $c_A < c_B$. The four key parameters in this framework are Δ (price tolerance), α (type share), t (spatial cost) and δ (differentiation and utility premium).

Within this framework, Δ (price sensitivity) determines the slope of consumers' response to $(p_B - p_A)$; α denotes the share of consumers who attach greater weight to non-price factors; service and trust contribute to δ , which may vary across consumer segments; and t (spatial and search cost) prevents consumers from fully following the lowest price in all contexts.

In this study's sample, δ_A and δ_B are both conceptually consistent with $\delta > 0$, but weighted differently across groups; t is positive but limited; and Δ is at a medium-high level. Therefore, the equilibrium outcome is not a price war close to cost; rather, the following steady state emerges, summarized in equation (3).

$$0 \leq p_A^* \leq p_B^*, \quad c_A < c_B \quad (3)$$

where p_A^* and p_B^* denote the equilibrium prices of bazaar vendors (A) and supermarkets (B), respectively, and $c_A < c_B$ reflects the assumed cost asymmetry. In the highly educated group with strong service preferences, $p_B^* - p_A^* > 0$ is sustainable; in the price-oriented group, low-priced vendors remain competitive; both parties can maintain positive profits without the need to clear the market through extreme price reductions.

The Bertrand adaptation is validated here—price sensitivity and loyalty coexist, and bazaar transactions are embedded in human relationships rather than single-price competition. The Hotelling adaptation shows that consumers are not primarily convenience-driven, but rather enjoy the process of comparing and "browsing", and that the attractiveness of the bazaar derives from the experience rather than from distance. Overall, the study shows that price sensitivity is present but not deterministic (Δ medium-high), trust and service are significant (α high), and spatial and search cost is positive but limited ($t > 0$). As a result, bazaars and supermarkets are not zero-sum substitutes, but rather a pattern of segmented coexistence between low-priced vendors and service-oriented supermarkets under a sub-population structure, where both parties can achieve long-term operation through differentiation and segmented pricing.

5.4. Estimation results

5.4.1. Baseline specification and subgroup overview

The baseline specification (Model 1) regresses CHOOSE_VENDOR on the five behavioral indices using ordinary least squares with the full sample ($N = 90$). Four additional subgroup specifications probe heterogeneity by education (Model 2, $\leq$ college; Model 3, bachelor and above) and by age (Model 4, high-age; Model 5, low-age). Table 3 reports overall fit across all five specifications, and Table 4 reports the corresponding coefficient estimates.

Table 3. Overall model fit across baseline and subgroup specifications

Indicator	Model 1 (Full)	Model 2 (Low-edu)	Model 3 (High-edu)	Model 4 (High-age)	Model 5 (Low-age)
Sample size (N)	90	59	31	75	15
R ²	0.623	0.701	0.677	0.663	0.661
Adjusted R ²	0.596	0.667	0.596	0.634	0.406
F-statistic	22.84	20.35	8.38	22.34	2.60
Prob(F)	< 0.001	< 0.001	< 0.001	< 0.001	0.106

Table 3. (continued)

Durbin–Watson	1.97	2.04	1.86	2.30	1.95
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All full-sample and education- and high-age specifications attain R^2 above 0.62 with the F-test joint-significant at the 0.1% level; only the low-age subgroup ($N = 15$) is statistically marginal, and its coefficients should be interpreted with caution. Durbin–Watson statistics near 2 indicate no serious autocorrelation.

Table 4. Regression coefficients across baseline and subgroup specifications (DV: CHOOSE_VENDOR; OLS)

Variable	Model 1 (Full)	Model 2 (Low-edu)	Model 3 (High-edu)	Model 4 (High-age)	Model 5 (Low-age)
Constant	3.083***	1.610	4.832**	2.440**	10.395*
DELTA_SCORE	0.847***	1.118***	0.729***	0.930***	0.614†
PRICE_PRIORITY	−0.368***	−0.216*	−0.453**	−0.308***	−0.900*
TRUST_SCORE	−0.241*	−0.396***	−0.432†	−0.216*	−0.833†
SPACE_COST	0.045	−0.036	0.041	0.051	−0.683
SEARCH_BEHAVIOR	0.022	0.121	0.048	0.027	0.144
FIRST_SEEN_BUY	−0.178*	−0.060	−0.294†	−0.161*	−0.733†

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed).

In the baseline (Model 1), price-related variables show the strongest effects—DELTA_SCORE has a positive coefficient of 0.847 ($p < 0.001$), indicating that consumers more sensitive to small price differences are more likely to choose bazaar vendors, while PRICE_PRIORITY has a negative coefficient of −0.368 ($p < 0.001$), indicating that consumers treating price as the primary factor lean toward supermarkets. TRUST_SCORE is also significant (−0.241, $p = 0.019$), suggesting trust-oriented consumers favor supermarkets in the full sample. FIRST_SEEN_BUY shows a small negative effect (−0.178, $p = 0.022$), while SPACE_COST and SEARCH_BEHAVIOR are insignificant. After controlling for trust, spatial and behavioral factors, price sensitivity and price priority therefore remain the main predictors of vendor choice.

5.4.2. Heterogeneity by education and age

Subgroup specifications reveal systematic heterogeneity along education and age dimensions (Table 5). By education, DELTA_SCORE is notably larger in the low-education subgroup (1.118 vs. 0.729), indicating that less-educated consumers respond more strongly to small price differences. In contrast, PRICE_PRIORITY has a larger absolute magnitude in the high-education subgroup (−0.453 vs. −0.216), suggesting that highly educated consumers who treat price as primary switch more decisively to supermarkets. Trust effects are stronger in the low-education subgroup (TRUST_SCORE = −0.396, $p < 0.001$), while in the high-education subgroup trust attains only marginal significance and an impulse-purchase effect emerges (FIRST_SEEN_BUY = −0.294, $p = 0.073$).

By age, the high-age subgroup closely follows the full-sample pattern but with sharper estimates; DELTA_SCORE (0.930, $p < 0.001$) and PRICE_PRIORITY (−0.308, $p < 0.001$) are still the main features, and TRUST_SCORE (−0.216) and FIRST_SEEN_BUY (−0.161) are both statistically significant at the 5% level. The low-age subgroup (Model 5) has the most extreme point estimates

-0.900 for PRICE_PRIORITY and -0.833 for TRUST_SCORE; however, due to a small sample size ($N = 15$), inference is limited, and only PRICE_PRIORITY remains statistically significant at the 5% level, while others are not. Space Cost and Search Behaviour are still insignificant for all subgroups; thus, spatial friction and explicit search do not play a leading role in the vendor selection decision for this sample.

Table 5. Heterogeneity summary across education and age subgroups

Mechanism	Education contrast	Age contrast	Interpretation
Price sensitivity (DELTA_SCORE)	Low-edu stronger (1.12 vs. 0.73)	High-age stronger (0.93 vs. 0.61†)	Less-educated and older consumers respond more reliably to price differences
Price prioritization (PRICE_PRIORITY)	High-edu greater magnitude (-0.45 vs. -0.22)	Low-age much stronger (-0.90 vs. -0.31)	Younger and more educated consumers switch decisively when treating price as primary
Trust orientation (TRUST_SCORE)	Low-edu significant (-0.40***)	High-age significant (-0.22*)	Trust matters more for less-educated and older consumers
Impulse purchase (FIRST_SEEN_BUY)	High-edu marginal (-0.29†)	High-age significant (-0.16*)	Bounded-rationality effect emerges in more deliberate decision-makers

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed).

Together, the results of heterogeneity show that the importance of price competition is consistent across all consumer groups, and the relative weight given to trust and impulse vs. deliberate decision-making changes predictably with education and age. Therefore, the bazaar is a differentiated market that has various categories of vendors and consumers.

6. Conclusion

6.1. Closing the loop

The Xinjiang Bazaar serves as the research subject, and this paper explores whether the traditional bazaar is a pure market for price competition. By building a theoretical framework that integrates price competition and spatial competition, and empirically observing the behaviour of consumers and merchants through questionnaire data, this paper finds that the competition mechanism of the bazaar in reality deviates significantly from the ideal model of "homogeneous products + complete rationality" in classical economics. According to the results of the study, consumers do not choose based solely on differences in prices but are also affected by trust relationships and spatial behaviour simultaneously; merchants' strategies for price and location also have multi-dimensional links. Therefore, the traditional market should not be viewed as a "price-war market", but rather as an all-encompassing system where price, relationship and space are intertwined. This conclusion responds to and further elaborates on the research questions presented in the introduction of this paper, thus achieving a closed loop of theory and empirical evidence.

6.2. Main conclusions

Based on the results of empirical analysis, this paper can draw the following three core conclusions. First, on the price dimension, consumers generally show high price sensitivity, indicating that price is still an important influencing factor in bazaar transactions. However, price does not constitute the

sole determinant, and consumers do not completely shift their purchases due to small price differences. Second, on the trust and service dimension, consumers are generally willing to accept relatively higher prices on the premise of familiarity with the vendor or a good service experience. This suggests that trust relationships and service quality can bring additional utility to traders, thus weakening the intensity of price competition to some extent. Third, on the spatial dimension, it is found that consumers do not show strong distance sensitivity, but rather prefer to compare and search, which implies that spatial cost is not a major constraint in the overall sample. Consumption in bazaars is therefore closer to an "active search–comparison–choice" behavioral pattern rather than a purely convenience-oriented one. Taken together, this paper concludes that the competitive structure of bazaars in Xinjiang is not single-dimensional, but rather a multidimensional equilibrium in which "price sensitivity (Δ), trust preference (α) and spatial cost (t) play a joint role". From the perspective of merchant analysis, the mainstream merchant strategy mix is closer to the competitive model of "price flexibility + space sensitivity". This strategy may help to maintain competitiveness in the short term, but at the overall level it is likely to lead to price-margin compression and increasing space cost, thus reducing the efficiency of market operation. By contrast, price rigidity or weak space sensitivity, although not mainstream, alleviates the problem of competitive homogenization to a certain extent.

6.3. Theoretical contribution

In the theory of price competition, the classical model usually assumes that products are homogeneous and that there are no non-price factors, which leads to the conclusion that prices tend to converge to marginal costs. However, this paper shows that in real markets, trust relationships and service experiences can constitute "non-price utility" that allows consumers to maintain a certain degree of loyalty in the face of price differences. This implies that there is a mechanism similar to "relational differentiation" in the market, which allows prices to be stabilized at a level higher than costs.

In terms of spatial competition theory, the classical model emphasizes the role of distance costs in market segmentation, while this paper finds that consumers tend to have a strong willingness to compare in the market, which makes the role of spatial constraints relatively limited. Space is more of a "search path" than a "segmentation boundary", which changes the logic of the traditional explanation of spatial competition. However, the difference in spatial awareness is asymmetric between consumers and merchants, which leads to a decrease in the efficiency of the market.

6.4. Limitations

Although there have been some attempts at theoretical integration and empirical investigation in this paper, there are still deficiencies. First, this paper mainly uses questionnaire surveys to obtain subjective perception data, and cannot observe objective indicators directly, such as actual prices, costs or market shares. Therefore, the research results are more indicative of behaviour and perceived structure rather than of market equilibrium. Second, only methods of correlation analysis and linear regression have been used in this paper; thus, the actual reasons for this phenomenon have not been revealed. Trust has not been systematically included in previous studies or the higher-order econometric models for analysis to determine whether it reduces price sensitivity and how space behaviour affects the strategy for setting prices. Some of the parameters in this paper (e.g., price sensitivity and spatial cost) are treated empirically in the model construction but are not precisely estimated; thus, the explanatory power of the model is limited to a certain extent. Third, in terms of

sample scope, this study has selected a particular regional marketplace, and therefore, the conclusions may be difficult to apply elsewhere or to other types of markets. Another reason for the above results may be the uneven distribution of age and education in the sample structure.

6.5. Implications

There are also several implications from the above results. First, traditional bazaars' competitive advantages lie in both low prices and their placement within social relations and cultural networks; thus, modernisation policies should not pursue "supermarketisation" or "e-commerce substitution" as ends in themselves but rather safeguard the trust mechanisms and social functions provided by bazaars. Second, in terms of regional development and poverty alleviation, particularly in areas with a low level of digitalisation or an older population, bazaars provide low-threshold employment and income sources for disadvantaged groups that are difficult to substitute. Third, by analyzing the heterogeneity of individual merchants, sustained differentiation through high-quality services and the cultivation of long-term customer relationships are more feasible than price competition in segment-specific scenarios with different consumer groups and price elasticities. Together, the above reasons indicate that the bazaars should be regarded as different types of socially integrated markets and included in the modernisation plan.

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